Lab Procedure for Simulink
Self-Localization

Setup

1. It is recommended that you review Lab 1 – Application Guide before starting this lab.
2. Launch Quanser Interactive Labs, scroll to the "QBot Platform" menu item and then select the "Warehouse" world.
3. Open the Simulink Model localization.slx, as shown in Figure 1. Loading the Simulink model also triggers the setup script for the Quanser Interactive Labs virtual environment. When the script is run successfully, it spawns the QBot and other elements within the virtual environment. Open the camera menu by pressing the 3 horizontal lines and set the Lock camera location to the "On" position.

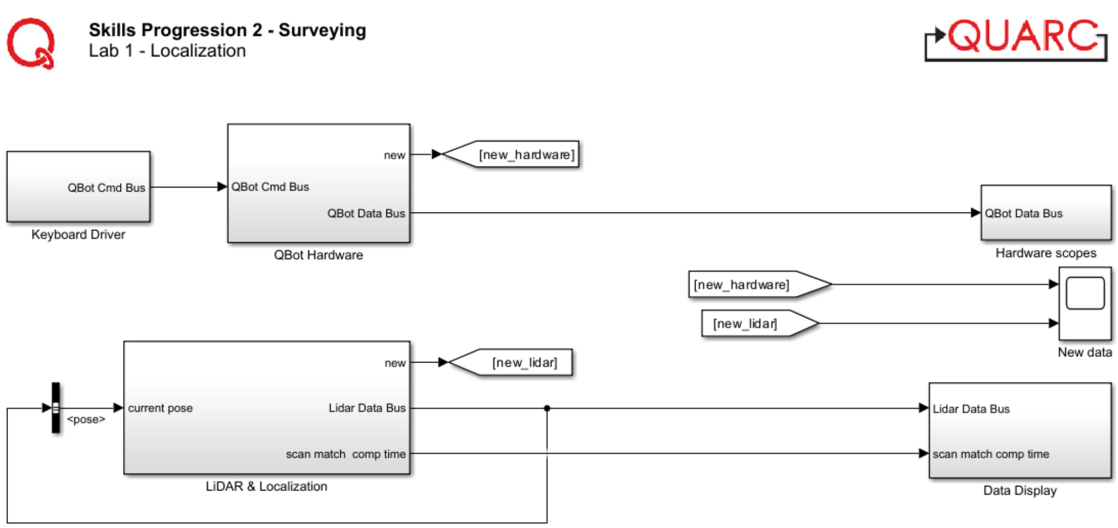

Figure 1: Localization Simulink Model

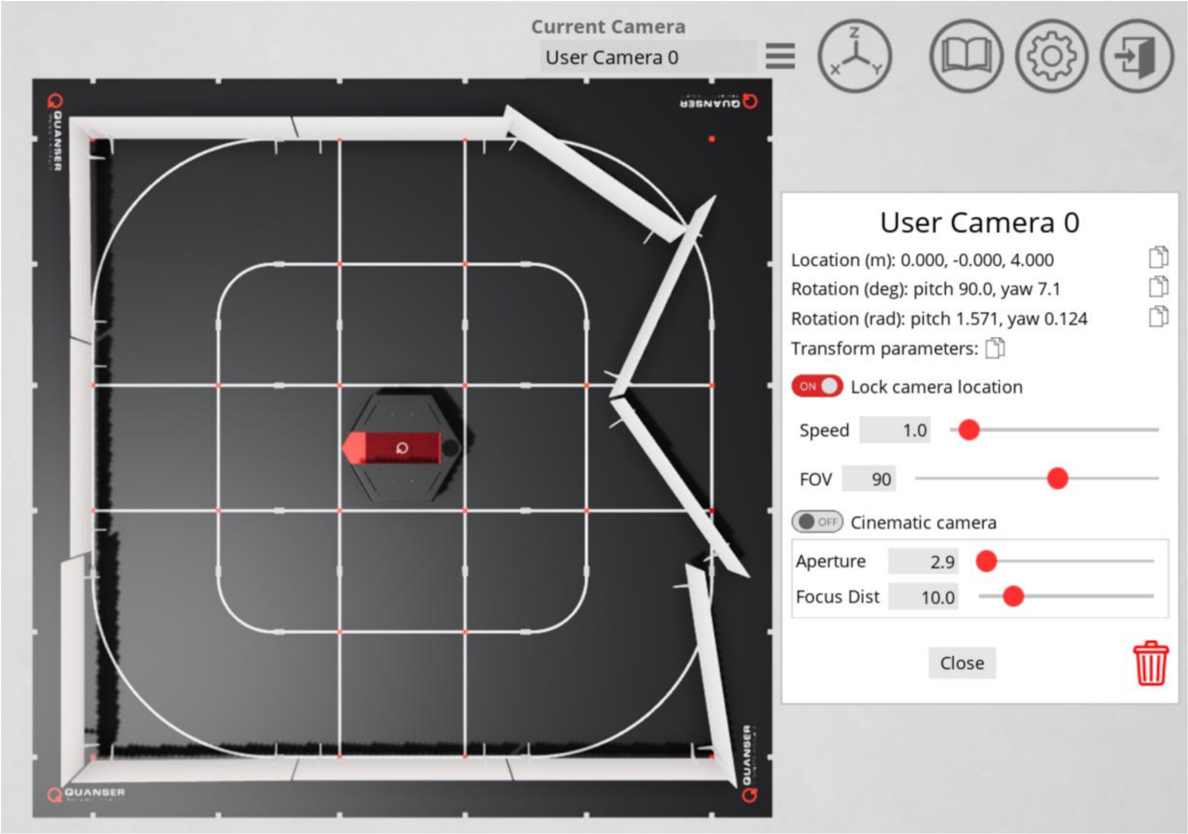

Figure 2. Successful set up of the Quanser Interactive Lab Workspace

Introduction to Scan Matching

1. Consider a situation, shown in Figure 2, where the QBot:

    a. Begins at location A
    b. Drives forward
    c. Stops at location B

    Suppose a LiDAR scan is taken at each location, giving you scan A and scan B. Describe how you would relate the data points in scan B to the points in scan A.

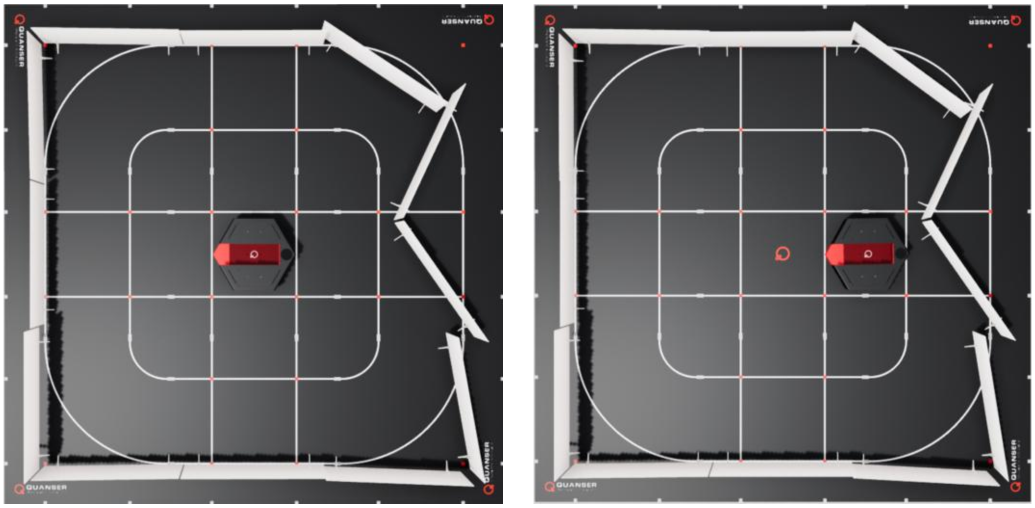

Figure 3. Top view of QBot in location A (left) and location B (right)

2. Consider a situation, shown in Figure 3 below, where the QBot:

    a. Begins facing direction C
    b. Rotates 90 degrees counterclockwise
    c. Stops rotating, so it is now facing direction D

    Suppose a LiDAR scan is taken at each location, giving you scan C and scan D. Describe how you would relate the data points in scan D to the points in scan C. What is the added challenge for rotation compared to translation?

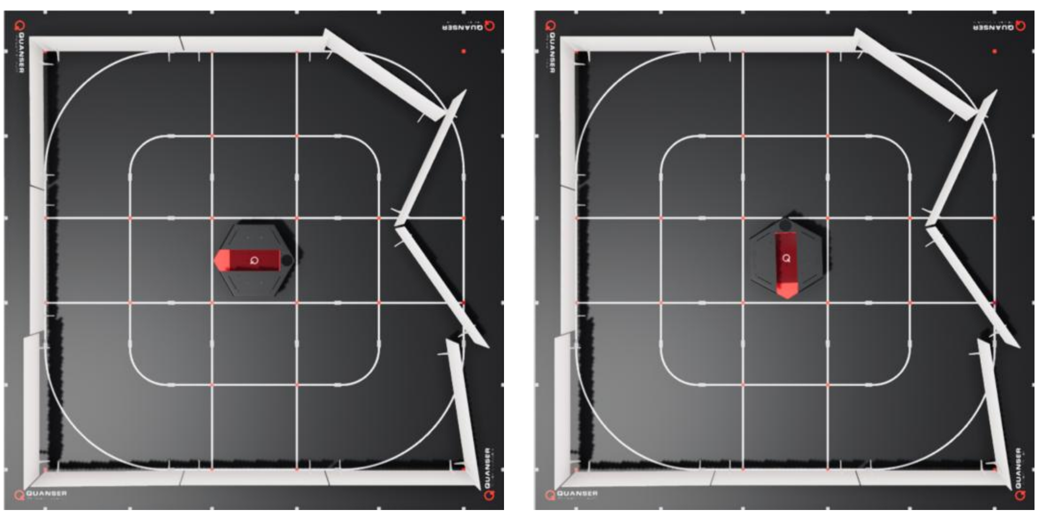

Figure 4. Top view of QBot facing direction C (left) and direction D (right)

Setting up LiDAR Scan Match

3. In this lab you will be exploring LiDAR-based localization using the Scan Match algorithm. In the Simulink model localization.slx, open the **LiDAR & Localization** subsystem. This subsystem will read data from the LiDAR sensor and perform scan match to estimate the position of the QBot. Open the **Subsample and Correct LiDAR** subsystem, then open the function block **correct_lidar** and complete the function with your code from Lab 4: Obstacle Detection in Skills Progression 1. Recall that this code corrects the raw headings so that 0 corresponds to the front of the QBot. If needed, review the lab procedure from Lab 4: Obstacle Detection.

4. Return to the **LiDAR & Localization** subsystem. Next, we will set up the inputs to the Scan Match block to estimate the position of the QBot using the subsampled LiDAR scan data. Open the **Run & Time Scan Match** subsystem. This subsystem runs and measures the computation time of the LIDAR Scan Match block using a function-call subsystem, shown in Figure 5 below.

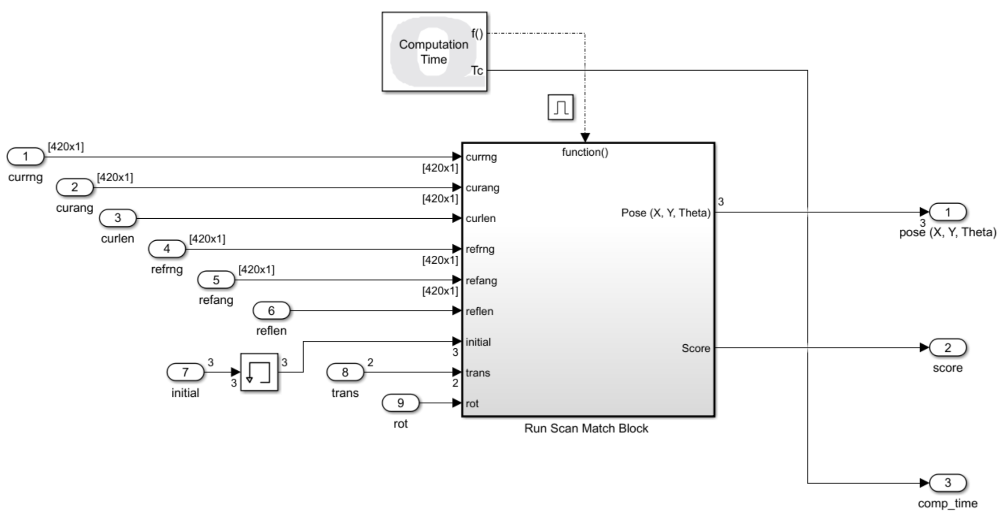

Figure 5. Inside of Run & Time Scan Match subsystem

Open the **Run Scan Match Block** subsystem. The **LIDAR Scan Match** block matches a reference LiDAR scan to the current LiDAR scan, performing a grid matching algorithm to determine the translation and rotation of the current scan relative to the reference. Right-click on this block and click on "help" to open the documentation. You can refer to this page for descriptions of input and output ports as we complete the rest of this lab.

5. Return to the **LiDAR & Localization** subsystem. In this lab, we will find the position of the QBot relative to its initial position, by using scan match to determine the transformation between its current scan and a saved reference scan from its initial position. First, connect the **correct_lidar** outputs to the **Run Scan Match Block** input ports for the current scan.

6. Next, we will complete the connections to the triggered **Freeze Scan** subsystem, which is set up to run with a rising edge trigger occurring when at least 3 seconds have elapsed. A triggered subsystem holds its outputs at the last value between triggers. Open the subsystem, where you can see that each input port connects directly to an output port. Return to the **Lidar & Localization** subsystem. Complete the connections to the **Freeze Scan** subsystem so it saves a copy of the current scan when triggered and passes the scan to the **Run Scan Match Block** subsystem.

7. The output of the **Lidar & Localization** subsystem is a data bus that contains current scan data, reference scan data, and the estimated pose output from the Scan Match Block. Complete the connections to the bus creator shown below: change figure to student version which will include dotted lines

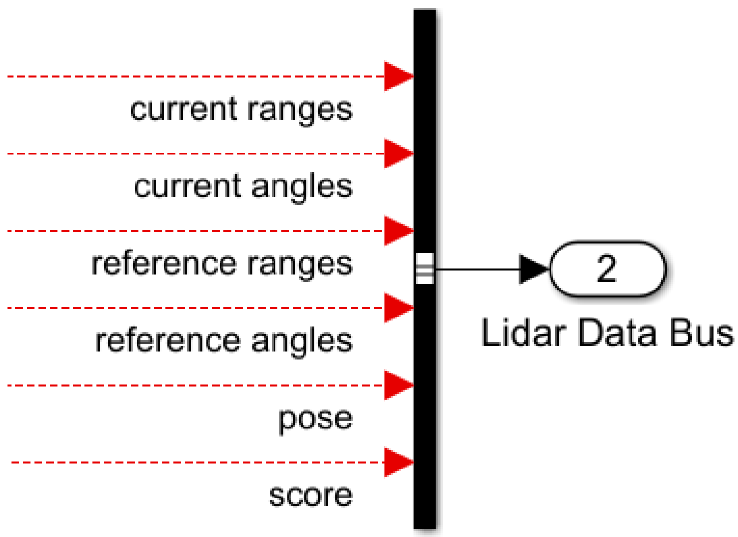

Figure 6. LiDAR Data Bus inputs

8. Return to the main Simulink model and note that the pose is selected from the Lidar Data Bus and then passed back to the **Lidar & Localization** subsystem. Open the Lidar & Localization subsystem again. The **LIDAR Scan Match** block starts the grid search at an initial position passed to the **initial** input. Connect this input to use the previous estimated position as the starting point for the current search.

9. The connections for the last two inputs are completed for you. The input **trans**, set to **[3, 3]**, specifies the extent of the translation search, in m, in the x and y directions. The input **rot** sets the angular search range in radians and is set to 2π.

10. Return to the main Simulink model. Run the model by clicking the Run button under the Simulation Tab. In Quanser Interactive Labs, ensure that the “Lock camera location” toggle switch is set to the “on” position to fix the camera location.

11. In the main Simulink model, open the **Data Display** subsystem. Open the Current Scan display, which displays the LiDAR scan data that is passed into the **currng** input of the **LIDAR Scan Match** block. Open the Current Pose display, which plots the current estimated position of the QBot relative to the reference LiDAR scan.

12. To drive the QBot using your keyboard, press and hold the space bar to arm the robot. Keep the space bar pressed as you drive the QBot using the movement keys. To stop the QBot, disarm it by releasing the space bar. The movement keys are as follows:

    a. Press the "A" and "D" keys to turn the QBot body

    b. Press the "I" and "K" keys to drive the Qbot forward and back

    Note that this lab does not include obstacle detection functionality. Drive the QBot around while observing the Current Scan and Current Pose displays. Which heading corresponds to the forward movement direction of the QBot?

13. Open Data Scopes within the **Data Display** subsystem, which opens three different plots. Drive the QBot around while observing these scopes. The Estimated Pose plot displays the x, y, and theta values of the current position estimate. Which axis corresponds to the forward movement direction of the QBot? The Score plot displays a non-normalized confidence for the match. The Timing scope displays sample and computation times for the model. The Scan Match Computation Time trace displays the output of the Computation Time block inside the Run and Time Scan Match subsystem (Figure 5). Stop the code using the Stop button under the Simulation Tab.

Exploring Parameters of LiDAR Scan Match

14. In this section, we will examine how scan match parameters impact performance. Open the following subsystems: LiDAR and Localization > Run and Time Scan Match > Run Scan Match Block.

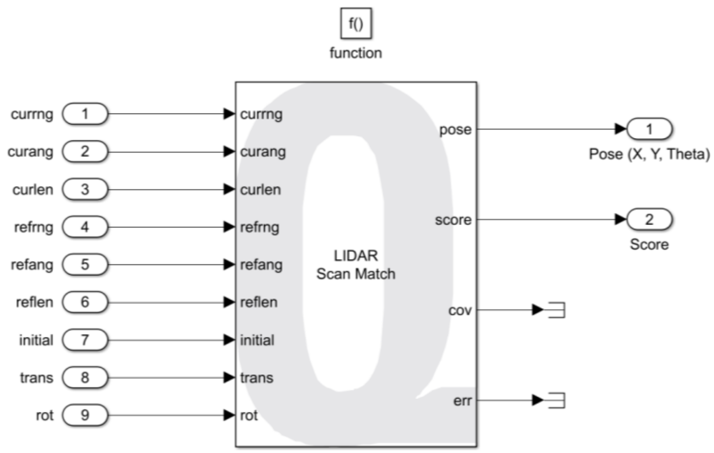

Figure 7. Inside of Run Scan Match Block subsystem

Click on the LIDAR Scan Match block, shown in Figure 7, to open block parameters. The **resolution** parameter, currently set to 10, sets the number of grid cells per meter. Increase the resolution to 20 and run the code. Drive the QBot around the map while observing the Estimated Pose, Score and Timing scopes. How does changing the resolution impact performance? If needed, increase the resolution by increments of 5 until you observe a significant effect. Stop your code.

15. Set the resolution back to 10. Inside the Lidar & Localization subsystem, change the constant connected to the **trans** input of the Scan Match Block to `[1 1]`. Run the code and drive the QBot around while observing the scopes. How is performance affected by changing the translation search range?

16. Stop the Simulink model when complete by pressing the "U" key. Ensure that you save a copy of your completed files for review later. Close Quanser Interactive Labs.